UNIFIED MENTOR PROJECT REPORT

Predictive Forecasting of Care Load & Placement Demand

A Time-Series Forecasting and Decision-Support Prototype for the U.S. Unaccompanied Children Program

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Streamlit dashboard: Live application link

This report documents an educational forecasting prototype developed against the Unified Mentor project brief. It is not an official HHS analysis or an operationally validated capacity-management system.

Executive Summary for Stakeholders

The Unified Mentor project brief asks for forward-looking estimates of the number of children in HHS care, an assessment of transfer-versus-discharge pressure, short-term discharge-demand estimates, model comparisons, uncertainty summaries, and a dashboard that could support proactive planning. This report evaluates the implemented notebook and its available outputs against that scope.

What the analysis found. The supplied CSV contained 1,170 raw rows, of which 450 were completely blank. Cleaning retained 720 dated observations from 12 January 2023 through 21 December 2025. Those observations span 1,075 calendar days, leaving 355 dates without records; no Saturday records appear in the extract. Six models were evaluated using seven rolling forecast origins and a 30-day horizon. SARIMA achieved the lowest MAE in each reported lead band.

Headline care-load forecasting results from the supplied notebook

Lead band	Targets scored	MAE	RMSE	Derived MAPE*
1–7 days	30	18.47	36.78	0.78%
8–14 days	29	42.95	67.16	1.87%
15–30 days	73	103.73	132.82	4.57%

**Derived from the notebook's custom score, Forecast accuracy* $= 100(1 - \text{MAPE})$*. It is a standard MAPE conversion, not a universal probability that a prediction is correct.*

What is only partially complete. Discharge demand is represented by a recent observed-record baseline of 9.6 children per reported day, with a seven-observation rolling baseline MAE of 5.3. This is a useful reference level but is not a fully benchmarked calendar-day discharge forecasting system. Empirical error bands are produced, but their future coverage is not calibrated. The analysis does not calculate capacity-breach probability or surge lead time because no approved capacity threshold was supplied. A forecast-stability index was not defined or computed.

Operational interpretation. The final SARIMA forecast in the notebook is dated 22 December 2025 through 20 January 2026. That horizon has passed. It is reported as a historical experiment only. The data snapshot also predates the latest data listed by the public HHS dataset source. Before any current planning use, refresh the authorized source data, rerun the complete evaluation, validate uncertainty coverage, and obtain an approved capacity measure.

Submission links. The project repository supplied by the author is PREDICTIVE-FORECASTING-CARE-LOAD-PLACEMENT; the supplied Streamlit application is available here. The hosted app could not be independently inspected in this review environment, so the report does not claim that every required dashboard control was verified.

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Abstract

Operational planning for the U.S. Unaccompanied Alien Children (UAC) Program requires anticipating changes in the number of children in the care system, rather than relying exclusively on retrospective dashboards. This project develops an exploratory forecasting pipeline for the daily HHS care population and a preliminary baseline for discharges, using a supplied extract of program records. After removing fully blank rows, the dataset contains 720 dated observations from 12 January 2023 through 21 December 2025. The observed dates span 1,075 calendar days, leaving 355 calendar dates without records; Saturday observations are absent. The primary target is the number of children in HHS care, treated as a stock variable. Discharges and transfers from CBP to HHS are treated as reported flow measures. Six forecasting approaches are compared: naïve persistence, a seven-day moving-average baseline, additive damped-trend exponential smoothing with weekly seasonality, SARIMA, random forest regression, and gradient boosting regression. Evaluation uses seven rolling forecast origins, each forecasting 30 calendar days, and scores only dates with observed HHS care-load values. SARIMA achieves the lowest mean absolute error (MAE) in all three lead-time bands: 18.47 children for days 1–7, 42.95 for days 8–14, and 103.73 for days 15–30. The persistence baseline MAEs are 31.70, 62.59, and 119.85, respectively. A final 30-day SARIMA point forecast from the historical data cutoff begins at 2,485 children and reaches approximately 2,557 by 20 January 2026. Because these forecast dates have passed, they are reported only as historical experimental output, not as current estimates. Discharge demand is summarized using a trailing observed-record baseline rather than presented as a validated daily placement forecast. The study highlights how reporting gaps, interpolated stock values, limited rolling origins, uncertain weekly seasonality, and the absence of approved capacity thresholds constrain operational interpretation. The pipeline is therefore a reproducible analytical prototype, not a validated capacity-management system.

Keywords: time-series forecasting; child welfare planning; HHS care load; placement demand; SARIMA; machine learning; walk-forward validation; missing data; operational early warning

1 Introduction

Organizations that deliver time-sensitive care must plan resources before demand materializes. In the U.S. Unaccompanied Alien Children Program, changes in the number of children entering custody, being transferred between agencies, and leaving care can alter demand for accommodation, case management, health services, and placement coordination. A retrospective dashboard can describe these changes after they occur, but operational planning also needs estimates of what may happen over the coming days and weeks.

The scope and target deliverables follow the project brief supplied by Unified Mentor (Unified Mentor project brief, 2026). The dataset schema is consistent with the public HHS

Unaccompanied Children Program daily data described by HHS / ORR (U.S. Department of Health and Human Services, 2026). The objective of this project is to examine whether the supplied dated records can support short-horizon forecasts of the number of children in HHS care and provide a cautious baseline for reported discharge activity. A secondary objective is to compare simple benchmarks, classical time-series methods, and tree-based machine-learning methods under a time-ordered validation design. The project also describes how forecast ranges could be compared with approved capacity thresholds to support early warning.

The study is deliberately framed as an exploratory forecasting analysis. It does not estimate individual children's outcomes, recommend individual placements, or infer causal effects of policy. It uses aggregate time-series data and does not contain the facility-level capacity, staffing, acuity, or caseworker workload variables required to make a complete operational capacity model.

1.1 Research questions

The analysis addresses the following questions:

RQ1. How does the observed HHS care population vary over the study period, and what data-quality limitations affect interpretation?

RQ2. How accurately do persistence, moving-average, exponential-smoothing, SARIMA, random-forest, and gradient-boosting methods forecast HHS care load at 1–7, 8–14, and 15–30 calendar-day horizons?

RQ3. What can a recent observed-discharge baseline say about short-term discharge activity when reporting dates are incomplete?

RQ4. What additional information is required to turn a care-load forecast into a defensible operational early-warning system?

1.2 Contributions

The practical contributions of the project are:

- a reproducible preprocessing workflow that distinguishes missing observations from genuine zero counts;
- a stock-versus-flow framing of the program variables;
- a comparison of six forecasting approaches using rolling-origin evaluation;
- a horizon-specific comparison of forecast error;
- a preliminary observed-day discharge baseline and a net-flow pressure indicator; and

- an explicit statement of the limits that prevent the results from being interpreted as validated operational alerts.

2 Alignment with Project Requirements

The table below distinguishes functionality demonstrated by the supplied notebook from project goals that remain partial or unverified. This distinction matters because the brief describes an operational decision-support objective, while the available experiment is an exploratory analytical prototype.

Table 1: Requirement traceability against the Unified Mentor project brief.

Brief requirement	**Status**	**Evidence and limitation**
Forecast children in HHS care	Implemented experimentally	Six forecasting approaches are compared; SARIMA is selected by short-horizon MAE and a historical 30-day forecast is generated. The notebook cutoff is 21 December 2025, so results are not current forecasts.
Time-series cleaning and continuity	Partial	Dates are parsed and a daily index is created. HHS stock values are linearly interpolated for model inputs; 355 dates are missing, and Saturday records are absent. Flows are not imputed as zero.
Trend and seasonal decomposition	Implemented descriptively	An additive decomposition with weekly period is plotted. Missing Saturdays mean the weekly component may partly reflect the reporting schedule.
Lag and rolling feature engineering	Implemented for ML models	Features include lags 1, 7 and 14; rolling means and standard deviations over 7 and 14 days; weekday, month and day-of-year variables.
Strict time-based split and walk-forward validation	Implemented with limitations	Seven rolling origins, 30-day forecasts and 127 observed validation targets are reported. Origins are few and validation windows overlap, limiting claims about robustness.
Baseline, statistical and ML model comparison	Implemented	Naive persistence, 7-day moving average, exponential smoothing, SARIMA, random forest and gradient boosting are evaluated using MAE and RMSE.
Predict discharge / placement demand	Partial	A trailing reported-observation baseline is evaluated, but no dedicated multi-model discharge comparison is reported. Missing dates prevent interpreting the baseline as a complete calendar-day placement forecast.

Brief requirement	**Status**	**Evidence and limitation**
Quantify forecast uncertainty	Partial	Empirical bands based on error quantiles are plotted. Coverage has not been calibrated on a sufficiently large independent test set, so these are not guaranteed 90% prediction intervals.
Estimate imbalance between transfer and discharge flows	Partial	The indicator $P_t = T_t - D_t$ is calculated on reported dates. It is not an exact change in HHS stock because timing, missing dates and other movements are not reconciled.
Capacity-breach probability	Not calculated	The input contains no approved capacity threshold or staffed-capacity series. A threshold and an agreed operational definition are required before this KPI can be estimated.
Forecast accuracy (Surge lead time	Not calculated	No validated surge definition or operational threshold is available, so lead time before a surge cannot be estimated defensibly.
Forecast stability index / model robustness	Not calculated / limited evidence	No stability-index formula is defined. Seven rolling origins provide only preliminary evidence, not a long-term robustness assessment.
Streamlit application and dashboard requirements	Link provided; live controls unverified	The brief's target modules are a future care-load chart, discharge-demand panel, model comparison and uncertainty view, plus horizon/model/scenario controls. The supplied GitHub and Streamlit URLs are included, but their live contents could not be independently inspected in this review environment.
Executive summary for stakeholders	Included	A stakeholder-oriented summary is provided at the beginning of this report and the quantitative findings are kept separate from operational claims.

3 Program Context and Analytical Framing

The source dataset describes measures associated with the UAC Program. The analysis is submitted as an applied project deliverable in response to the project specification provided by Unified Mentor, rather than as an official HHS analysis. The columns include children apprehended and placed in CBP custody, children in CBP custody, children transferred out of CBP custody, children in HHS care, and children discharged from HHS care. In this report, the term *HHS care load* refers to the recorded count of children in HHS care.

For modeling, it is essential to distinguish a *stock* from a *flow*. A stock is a quantity measured at a point in time; a flow is a quantity accumulated over a reporting interval.

Let:

$$H_t = \text{number of children in HHS care at time } t, \tag{1}$$
$$T_t = \text{reported transfers into HHS during period } t, \tag{2}$$
$$D_t = \text{reported discharges from HHS during period } t. \tag{3}$$

Under aligned definitions and reporting periods, a simplified balance relationship is

$$H_t = H_{t-1} + T_t - D_t + U_t, \tag{4}$$

where U_t represents other movements, timing differences, adjustments, or unmeasured components. If all movements are complete and precisely aligned, U_t may be zero. The supplied dataset does not establish that this condition holds for every record. Accordingly, the difference $T_t - D_t$ is treated as a *net-flow pressure indicator*, not as an exact daily change in H_t.

The operational motivation is straightforward: if incoming transfers persistently exceed discharges, the active population may rise. However, an increase in care load is not synonymous with overcrowding. Overcrowding is a capacity relationship and requires verified measures of available, staffed, and appropriate care capacity.

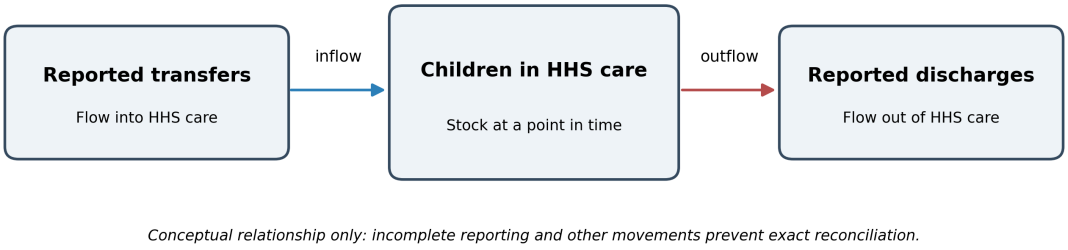


Figure 1: Conceptual stock-and-flow relationship among reported transfers, the HHS care population, and reported discharges. This diagram is illustrative; incomplete reporting and other movements prevent exact reconciliation from the supplied series alone.

4 Dataset and Data Quality

4.1 Variables

The input file is named `HHS_Unaccompanied_Alien_Children_Program.csv`. The source column names and analytical roles are summarized in table 2.

Table 2: Source variables and their roles in the analysis.

Source variable	Analytical label	Role and interpretation
Date	Date	Reporting date and time index.
Children apprehended and placed in CBP custody*	`cbp_intake`	Upstream intake measure; used for context, not as the primary forecast target.
Children in CBP custody	`cbp_care`	Upstream stock measure.
Children transferred out of CBP custody	`transfers_to_hhs`	Reported flow indicator toward HHS care.
Children in HHS Care	`hhs_care`	Primary target: active HHS care-load stock.
Children discharged from HHS Care	`discharges`	Secondary target: reported discharge flow.

4.2 Cleaning and coverage

The source contained 1,170 raw rows, including 450 fully blank rows. Removing fully blank rows left 720 dated observations. Dates range from 12 January 2023 to 21 December 2025. The inclusive calendar span contains 1,075 days, so 355 dates have no source record. The longest gap between adjacent records is 10 days. Approximately 77.6% of adjacent observations are one calendar day apart. The weekday summary contains no Saturday observations.

These characteristics are central to the interpretation of results. A missing date does not establish that no child entered or left care. In particular, a missing discharge count must not be converted to zero without evidence that the reporting system uses zero to represent a confirmed zero-activity day. The flow series are therefore retained only on reported dates for descriptive and baseline calculations.

Table 3: Data coverage reported by the executed notebook.

Quantity	Value
Raw rows	1,170
Fully blank rows removed	450
Dated observations retained	720
First date	2023-01-12
Last date	2025-12-21
Inclusive calendar span	1,075 days
Missing calendar dates	355
Longest gap between records	10 days
Adjacent observations one day apart	Approximately 77.6%
Saturday observations	None in the extract

4.3 Parsing and validation

The workflow parses dates as datetime values, reads numeric values while accounting for thousands separators, renames the source columns to concise labels, sorts records chronologically, and checks for invalid dates, duplicate dates, and missing metric values among retained rows. Duplicate dates and unresolved missing metric values cause the pipeline to stop rather than silently merge or impute them.

The absence of missing values within retained rows does not mean that the time series is complete. Row-level completeness and calendar coverage are different properties: each retained row can be complete while hundreds of dates are absent.

4.4 Daily reindexing and interpolation policy

A continuous daily index is created from the first to the last observed date. Only the HHS care-load stock is interpolated in time for model-input construction. If observed values H_a and H_b occur at dates a and b, a linear interpolation at an intermediate date t is

$$\widetilde{H}_t = H_a + \frac{t-a}{b-a}(H_b - H_a), \qquad a < t < b, \tag{5}$$

where dates are expressed in consistent time units. This estimates a straight-line path between the observed endpoints. It does not recover the unknown actual daily path and can smooth real spikes or drops.

The notebook preserves an observed-value mask. Interpolated HHS values may be used as lag inputs to create a regular daily modeling series, but forecast scores are calculated only when an actual observed HHS target is available. Flow variables are not filled with zero and are not interpolated into purported daily totals.

This policy is a pragmatic modeling choice, not a claim that linear interpolation is statistically optimal. The impact of alternative missing-data strategies should be assessed in future work.

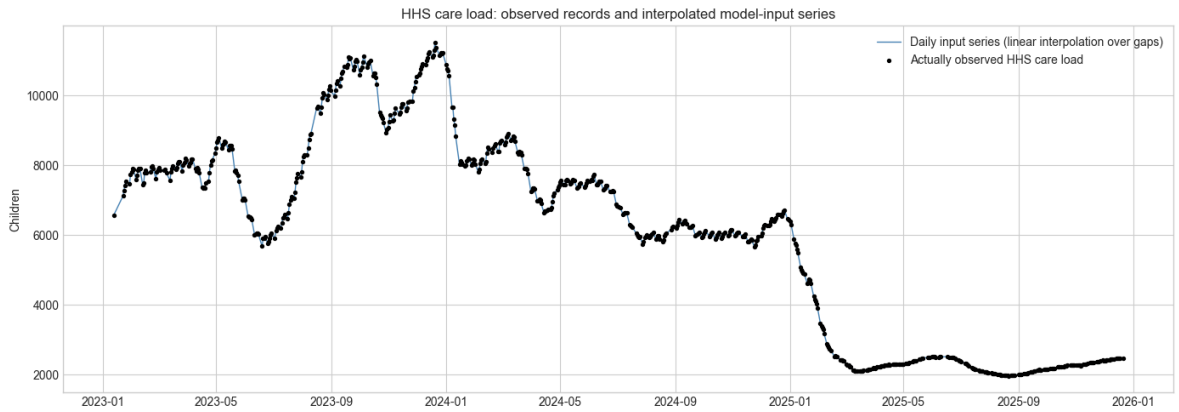


Figure 2: Observed HHS care-load records and the daily input series created by linear interpolation over reporting gaps. The interpolated line is a modeling construct, not a recovered measurement of the true daily population.

5 Exploratory Time-Series Analysis

5.1 Visual evidence in the supplied figures

The observed-measures plot shows the HHS care-load stock rising above 10,000 children during late 2023, declining through 2024, and dropping sharply in early 2025 to around 2,000 before a modest recovery toward the end of the displayed period. These large level changes imply that a single global trend would be inadequate and that model performance should be tested across distinct regimes. The transfer and discharge panels also show a marked reduction in reported flow counts during early 2025. The figures alone cannot determine whether this reflects a real operational change, a change in reporting, or both; external documentation is needed to distinguish these explanations.

The interpolation plot makes the modeling distinction visible: the line spans dates without source observations, while the point markers identify actually observed HHS values. The smoother daily path is useful for constructing lagged features and fitting a regular-grid model, but it must not be presented as observed daily data. The decomposition plot shows a nearly fixed-amplitude weekly component and structured residual bands in the later period. Given the absent Saturday records, this pattern may partly reflect the imposed regularization and should not be treated as independent evidence of a genuine weekly behavioral cycle.

The historical forecast plot begins at approximately 2,485 children and rises gradually toward approximately 2,557 by the end of its 30-day horizon. Its shaded band widens at longer lead times. The discharge plot shows a flat baseline near 9.6 children per reported day with a broad empirical band. Because the observed discharge values vary substantially and are not present for every calendar day, this baseline is best interpreted as a simple reference level, not a precise cumulative placement-demand estimate.

5.2 Trend, seasonality, and residual structure

The HHS stock series is examined using additive seasonal decomposition with a period of seven days. The additive representation is

$$Y_t = T_t + S_t + R_t, \tag{6}$$

where Y_t is the modeled series, T_t is its trend component, S_t is the seasonal component, and R_t is the residual component.

The decomposition is descriptive. It separates estimated components under the assumptions of the decomposition procedure; it does not establish a causal or stable weekly process. Since the source has no Saturday records, interpolated values supply the series on Saturdays. A weekly component may therefore reflect the reporting schedule as well as actual program behavior. The seasonal pattern must not be interpreted as reliable evidence of Saturday activity.

The notebook also summarizes reported transfers and discharges by observed weekday. Because Saturday records are absent and coverage is irregular, these summaries describe the available reports rather than a balanced sample of all calendar days.

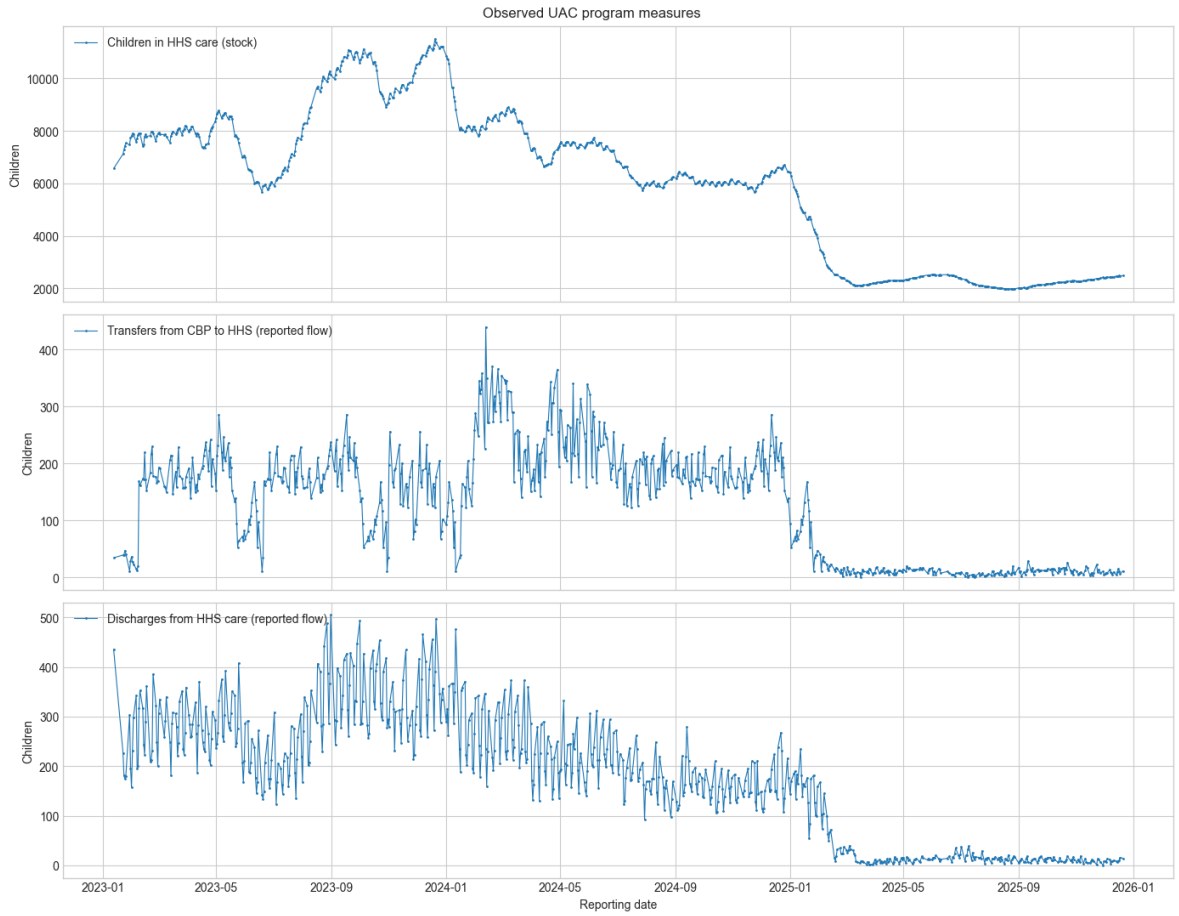


Figure 3: Observed UAC program measures in the supplied extract. The HHS care-load stock declines sharply in early 2025, while reported transfers and discharges also fall. The chart displays recorded observations; it does not establish that unreported dates represent zero activity.

6 Forecasting Methodology

6.1 Forecast target and horizon

The primary forecast target is H_t, the number of children in HHS care. Each evaluation origin generates a forecast for the next 30 calendar days. The evaluation groups forecast leads into:

$$\mathcal{H}_1 = \{1, \ldots, 7\},$$
$$\mathcal{H}_2 = \{8, \ldots, 14\},$$
$$\mathcal{H}_3 = \{15, \ldots, 30\}.$$

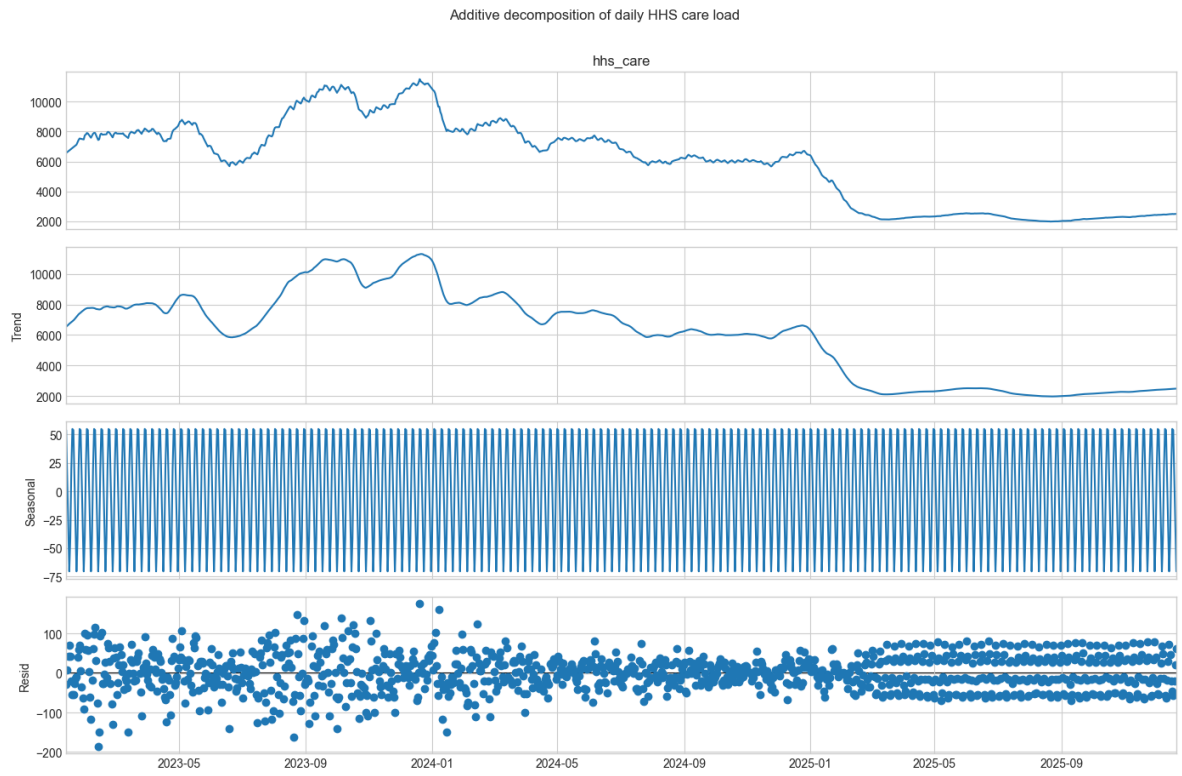


Figure 4: Additive decomposition of the daily HHS care-load modeling series. The estimated weekly component is mechanically regular and should be interpreted cautiously because the source contains no Saturday observations and interpolated values contribute to the daily series.

The distinction between calendar-day horizon and observed-record horizon is important. The HHS stock is modeled on a daily index, while actual target values are available only on recorded dates. A forecast can therefore exist for a date for which no observed value is available to score it.

6.2 Baseline 1: naïve persistence

The persistence baseline assumes the latest available care load remains unchanged:

$$\widehat{H}_{t+h|t} = H_t, \qquad h = 1, \ldots, 30. \tag{7}$$

This is a useful benchmark because it is simple, transparent, and often competitive for short horizons when a series changes slowly.

6.3 Baseline 2: seven-day moving average

The moving-average baseline uses the mean of the most recent seven daily values and repeats it over the forecast horizon:

$$\widehat{H}_{t+h|t} = \frac{1}{7} \sum_{j=0}^{6} \widetilde{H}_{t-j}, \qquad h = 1, \ldots, 30, \tag{8}$$

where $\widetilde{H}$ denotes the regularized modeling series. This reduces sensitivity to a single observation but does not dynamically extrapolate a trend.

6.4 Additive damped-trend exponential smoothing

The exponential-smoothing model uses an additive trend, a damped trend component, and additive seasonality with period $m = 7$. A common Holt–Winters formulation is

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + \phi b_{t-1}), \tag{9}$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)\phi b_{t-1}, \tag{10}$$

$$s_t = \gamma(y_t - \ell_{t-1} - \phi b_{t-1}) + (1-\gamma)s_{t-m}, \tag{11}$$

$$\widehat{y}_{t+h|t} = \ell_t + \left(\sum_{j=1}^{h} \phi^j\right) b_t + s_{t+h-m(k+1)}. \tag{12}$$

Here ℓ_t denotes the level, b_t the trend, s_t the seasonal component, α, β, γ are smoothing parameters, and ϕ is the damping parameter. The seasonal index in the forecast equation is selected by cycling through the estimated seasonal pattern. The exact state-update and initialization details are handled by the `statsmodels` implementation. The model is fitted with optimized parameters and estimated initialization.

The damped trend reduces the tendency of a linear trend to continue indefinitely at full strength. This can be useful for operational series whose short-term trends may not persist over long horizons.

6.5 SARIMA

The principal statistical model is SARIMA$(1,1,1) \times (1,1,1)_7$. In general, a seasonal autoregressive integrated moving-average model can be expressed as

$$\Phi_P(B^s)\,\phi_p(B)(1-B)^d(1-B^s)^D y_t = c + \Theta_Q(B^s)\,\theta_q(B)\epsilon_t, \tag{13}$$

where B is the backshift operator, $By_t = y_{t-1}$; p, d, q are the nonseasonal autoregressive, differencing, and moving-average orders; P, D, Q are the seasonal orders; s is the seasonal period; and ϵ_t is an innovation term. For this experiment, $(p,d,q) = (1,1,1)$, $(P,D,Q) = (1,1,1)$, and $s = 7$.

The model is estimated through the `SARIMAX` implementation in `statsmodels`, with stationarity and invertibility constraints relaxed and a maximum of 150 optimizer iterations. The model is fitted to the regularized daily HHS series available at each forecast origin. The seven-day seasonal period should be interpreted cautiously given the missing Saturdays and the broader gaps in source coverage.

6.6 Machine-learning features

The machine-learning models use lagged HHS values, rolling summaries, and calendar features. For each date t, the feature vector is

$$\mathbf{x}_t = \left[\widetilde{H}_{t-1}, \widetilde{H}_{t-7}, \widetilde{H}_{t-14}, \mu_{t,7}, \mu_{t,14}, \sigma_{t,7}, \sigma_{t,14}, w_t, m_t, d_t\right], \tag{14}$$

where w_t is day of week, m_t is month, and d_t is day of year. The rolling features are calculated from lagged values to avoid including the target date's HHS value in its own predictors.

The rolling mean and sample standard deviation are

$$\mu_{t,k} = \frac{1}{k}\sum_{j=1}^{k} \widetilde{H}_{t-j}, \tag{15}$$

$$\sigma_{t,k} = \sqrt{\frac{1}{k-1}\sum_{j=1}^{k}(\widetilde{H}_{t-j} - \mu_{t,k})^2}, \qquad k \in \{7, 14\}. \tag{16}$$

The training labels for the machine-learning models are restricted to actually observed HHS values. Interpolated values can enter the feature history but are not treated as observed training targets. This is intended to reduce the risk that artificial interpolated labels are mistaken for actual outcomes.

6.6.1 Random forest regression

Random forest regression averages the predictions from an ensemble of decision trees:

$$\widehat{H}_t = \frac{1}{M}\sum_{m=1}^{M} f_m(\mathbf{x}_t), \tag{17}$$

where M is the number of trees and f_m is the prediction of tree m. The notebook uses 250 trees, minimum leaf size 3, `max_features`=0.9, and a fixed random seed of 42.

6.6.2 Gradient boosting regression

Gradient boosting builds an additive ensemble sequentially:

$$F_M(\mathbf{x}) = F_0(\mathbf{x}) + \sum_{m=1}^{M} \nu\, f_m(\mathbf{x}), \tag{18}$$

where F_0 is the initial predictor, f_m is a fitted weak learner, and ν is the learning rate. The implementation uses 150 estimators, learning rate 0.03, maximum tree depth 2, Huber loss, and random seed 42.

6.7 Recursive multi-step forecasting

For the tree-based models, the forecast is recursive. The model predicts the next date, appends that prediction to the history, recalculates the lag and rolling features, and predicts the following date. If $\widehat{H}_{t+1}$ is predicted first, then subsequent feature vectors can depend on earlier predictions:

$$\widehat{H}_{t+h|t} = f\Big(\mathbf{x}_{t+h}\left(\widetilde{H}_{\le t}, \widehat{H}_{t+1|t}, \ldots, \widehat{H}_{t+h-1|t}\right)\Big). \tag{19}$$

Recursive forecasts can accumulate error because predicted values become inputs to later predictions. Performance at longer horizons must therefore be evaluated separately rather than inferred from one-step accuracy.

7 Time-Ordered Validation Design

7.1 Rolling-origin evaluation

Random train-test splitting is inappropriate for forecasting because it can train a model on observations later than those used to assess earlier predictions. The notebook instead evaluates the final 20% of observed HHS records using rolling forecast origins. Origins are separated by 28 calendar days. At each origin, the model receives only information available up to that date and generates a 30-day forecast. A forecast is scored only if the actual HHS value is observed on that future date.

The validation period begins on 14 May 2025. The executed run uses seven rolling origins. Forecast windows overlap, and some dates can contribute to more than one origin-specific error. The resulting records are useful for a preliminary model comparison but are not independent samples. The notebook reports 127 unique observed validation dates across the evaluation, with the number of scored forecast-origin/date pairs varying by horizon.

7.2 Error definitions

Let y_i denote an observed HHS care-load value and $\hat{y}_i$ its forecast. For n scored forecast cases, mean absolute error is

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \tag{20}$$

MAE is measured in children and is straightforward to interpret.

Root mean squared error is

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{21}$$

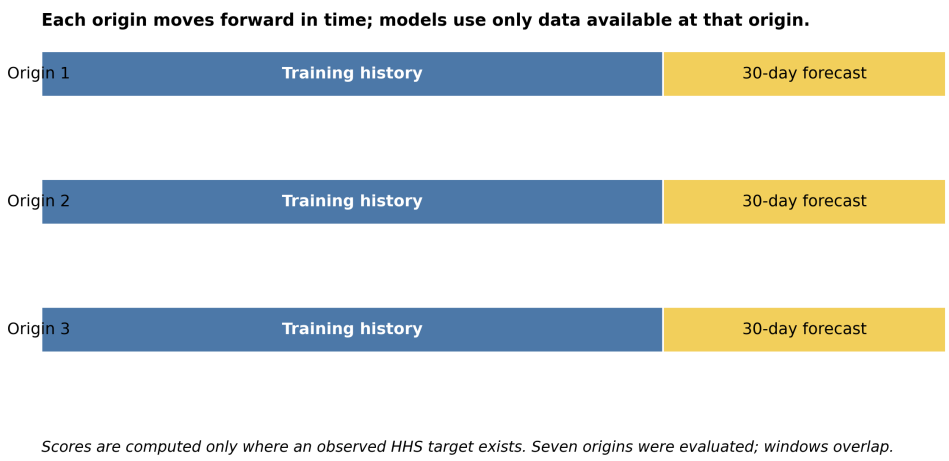


Figure 5: Schematic of rolling-origin validation. The training history advances through time, each origin forecasts the next 30 calendar days, and errors are scored only where an observed HHS target is available. The illustration is conceptual and not to scale.

Squaring the errors before averaging gives larger errors greater influence, making RMSE sensitive to occasional large misses.

Mean absolute percentage error is

$$\mathrm{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \tag{22}$$

for nonzero actual values. MAPE is undefined when $y_i = 0$ and can behave poorly near zero. In this dataset, the HHS stock is on a much larger scale than many possible daily flow counts. MAE and RMSE are therefore the primary metrics for operational interpretation.

The notebook also prints a quantity labelled "Forecast accuracy (%)" computed as

$$100\,(1 - \mathrm{MAPE}_{\mathrm{fraction}})\,, \tag{23}$$

with negative values clipped to zero. This is not a standard regression accuracy measure and should not be interpreted as the probability that a forecast is correct. The manuscript reports MAE and RMSE rather than treating this transformed percentage as a meaningful accuracy score.

7.3 Horizon-specific evaluation

Errors are grouped by forecast lead:

- short horizon: 1–7 days;
- medium horizon: 8–14 days;
- longer horizon: 15–30 days.

This is operationally motivated: forecasts that are useful for near-term staffing may be much less reliable for planning several weeks ahead.

8 Results

8.1 Care-load model comparison

The executed notebook reports the results in table 4. SARIMA has the lowest MAE among the six tested methods in each lead-time band. Its error increases with forecast lead, as expected for a multi-step forecast.

Table 4: Walk-forward forecast performance for HHS care load. Errors are measured in children.

Lead band	Model	n	MAE	RMSE	MAPE (%)
1–7 days	SARIMA	30	18.47	36.78	0.78
1–7 days	Naïve persistence	30	31.70	45.52	1.44
1–7 days	Random forest	30	39.85	54.15	1.82
1–7 days	7-day moving average	30	47.82	65.61	2.18
1–7 days	Exponential smoothing	30	62.62	67.91	2.87
1–7 days	Gradient boosting	30	143.81	175.21	6.83
8–14 days	SARIMA	29	42.95	67.16	1.87
8–14 days	Naïve persistence	29	62.59	77.90	2.88
8–14 days	Random forest	29	72.06	90.93	3.33
8–14 days	7-day moving average	29	77.21	97.59	3.56
8–14 days	Exponential smoothing	29	84.34	89.05	3.78
8–14 days	Gradient boosting	29	240.80	282.50	11.39
15–30 days	SARIMA	73	103.73	132.82	4.57
15–30 days	Exponential smoothing	73	111.97	138.30	4.95
15–30 days	Naïve persistence	73	119.85	134.97	5.41
15–30 days	Random forest	73	131.57	148.66	5.97
15–30 days	7-day moving average	73	135.03	152.75	6.12
15–30 days	Gradient boosting	73	315.69	355.18	14.75

MAPE values in this table are reconstructed from the notebook's reported score using MAPE = 100 − Forecast accuracy; the notebook itself drops the MAPE column after deriving that score. These values are therefore reported to the same two-decimal precision as the saved output.

For additional interpretation, the relative reduction in MAE compared with persistence is

$$\text{MAE reduction} = 100 \times \frac{\text{MAE}_{\text{persistence}} - \text{MAE}_{\text{model}}}{\text{MAE}_{\text{persistence}}}. \tag{24}$$

Using this measure, SARIMA reduces MAE by approximately 41.7% for days 1–7, 31.4% for days 8–14, and 13.5% for days 15–30 relative to the persistence baseline. This comparison is relative to one baseline and one validation setup; it does not establish universal superiority.

The results indicate that SARIMA captures temporal structure in this historical series more effectively than the other evaluated approaches. Gradient boosting has substantially larger errors in this run. Possible explanations include limited training data, the irregular reporting pattern, recursive error accumulation, and a mismatch between the fitted feature relationships and the underlying process. These explanations are hypotheses; the experiment does not isolate their causal contributions.

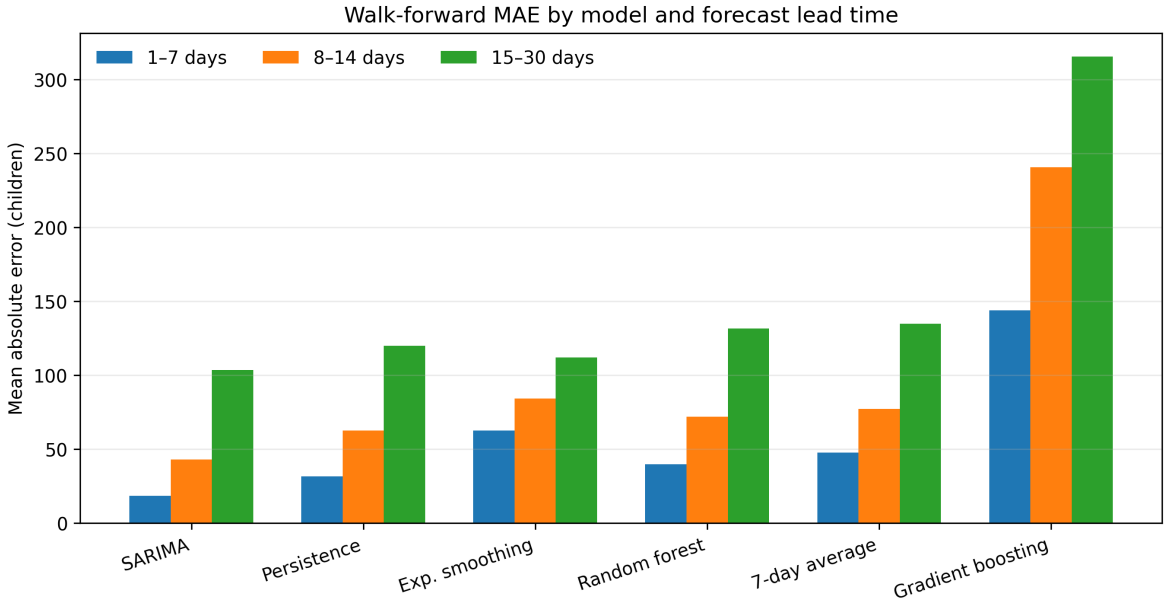


Figure 6: Mean absolute error (MAE) for six forecasting approaches across three lead-time bands, using the values reported by the executed notebook. Lower values indicate smaller absolute errors.

8.2 Historical 30-day care-load forecast

The final model is selected using the lowest 1–7-day walk-forward MAE and refitted using all observed HHS history available in the supplied extract. SARIMA is selected. Its forecast begins on 22 December 2025, immediately after the final observation, and extends through 20 January 2026.

The point forecast starts at 2,485 children and reaches approximately 2,557 children at the end of the 30-day horizon, a net increase of about 72 children. These are historical model outputs based on a data cutoff of 21 December 2025. They are not forecasts for the present date and must not be used for current resource decisions without refreshing the source and rerunning the analysis.

Table 5: Selected historical forecast checkpoints. Values are rounded to the nearest child.

Forecast date	Point forecast	Lower empirical band	Upper empirical band
2025-12-22	2,485	2,443	2,526
2025-12-23	2,487	2,445	2,528
2025-12-24	2,488	2,446	2,529
2025-12-28	2,500	2,459	2,542
2025-12-29	2,501	2,414	2,589
2026-01-04	2,517	2,430	2,605
2026-01-20	Approximately 2,557	Horizon-specific empirical band	Horizon-specific empirical band

8.3 Empirical error bands

For each lead-time band, the notebook calculates the 90th percentile of absolute walk-forward errors for the selected model. Let q_b be the empirical 90th percentile for lead band b. The displayed range is

$$L_{t+h} = \max(0, \widehat{H}_{t+h} - q_b), \tag{25}$$

$$U_{t+h} = \widehat{H}_{t+h} + q_b. \tag{26}$$

The lower bound is truncated at zero because a population count cannot be negative.

Although the notebook labels these ranges as empirical 90% bands, they are not calibrated 90% prediction intervals. The use of a 90th percentile of absolute errors does not guarantee 90% future coverage. Coverage must be assessed on representative held-out forecasts, ideally with enough independent forecast origins to evaluate stability. These bands should be described as empirical error bands until such calibration has been completed.

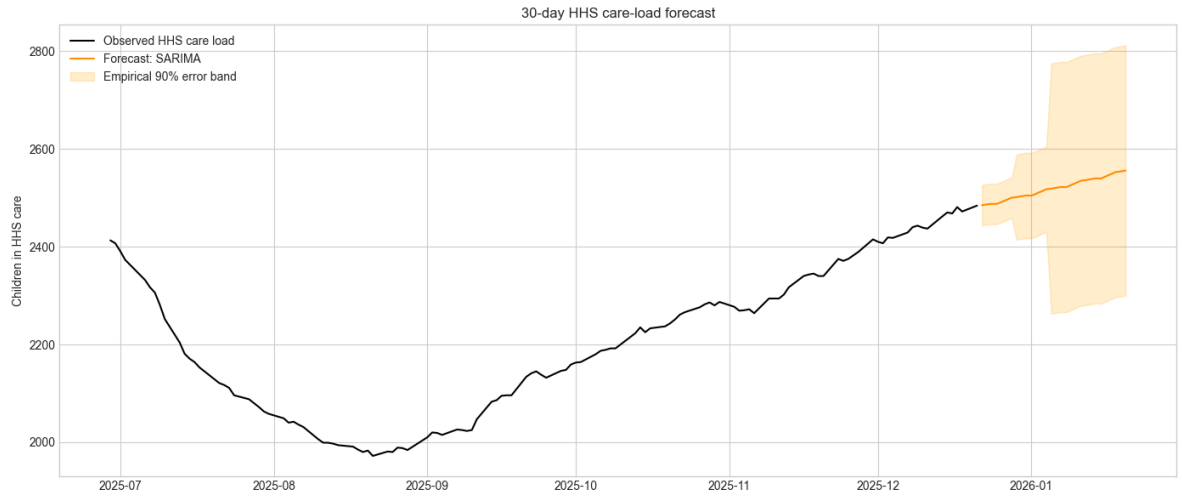


Figure 7: Thirty-day SARIMA forecast for HHS care load based on the historical cutoff. The plotted forecast begins after the final observed date in December 2025 and extends into January 2026. The shaded region is an empirical error band, not a calibrated 90% prediction interval.

8.4 Discharge-demand baseline

The discharge series is less suitable for a daily forecast because unreported dates cannot be interpreted as zero-discharge days. The notebook therefore calculates a recent observed-record baseline rather than claiming a fully validated calendar-day discharge model.

The mean of the last 14 reported discharge observations is 9.6 children per reported day. A trailing seven-reported-observation mean is evaluated at the same rolling-origin dates. The resulting baseline MAE is 5.3 children, calculated across 132 observed validation targets and seven origins.

The baseline forecast is constant over the next 30 calendar dates, and an empirical margin is derived from the 90th percentile of absolute validation errors. This is an indicative scenario under similar reporting conditions, not a validated estimate of the total number of discharges or placements over 30 calendar days. Multiplying 9.6 by 30 would implicitly assume a complete daily reporting process and a stable rate; neither assumption is established by the supplied data.

$$\widehat{D}_{t+h|t} = \frac{1}{14} \sum_{j=1}^{14} D_{r_j}, \tag{27}$$

where D_{r_j} denotes the jth most recent reported discharge observation, not necessarily the jth previous calendar day. For the evaluation baseline, the mean of the seven most recent reported observations is used:

$$\widehat{D}^{(7,\text{reported})}_{t+h|t} = \frac{1}{7} \sum_{j=1}^{7} D_{r_j}. \tag{28}$$

Table 6: Discharge and net-flow baseline findings from the executed notebook.

Measure	Reported result
Mean of last 14 reported discharge values	9.6 children per reported day
Trailing-seven-reported-observation baseline MAE	5.3 children
Observed discharge validation targets	132
Rolling origins for discharge evaluation	7
Recent transfer baseline	8.6 children per reported day
Recent net-flow baseline	Approximately −1.1 children per reported day
Mean observed net-flow pressure across reported dates	−44.7 children

8.5 Net-flow pressure

The notebook defines the reported net-flow pressure as

$$P_t = T_t - D_t. \tag{29}$$

A positive value indicates that reported transfers exceed reported discharges on that date; a negative value indicates the reverse. The recent mean transfer baseline is 8.6 and the recent mean discharge baseline is 9.6 children per reported day, giving a displayed recent pressure of approximately -1.1 after rounding. The mean pressure across all reported dates is -44.7 children.

The difference between the recent baseline and the historical mean shows that the selected time window materially affects the summary. The pressure metric must not be equated with the exact change in HHS care load because the source does not establish complete, aligned daily reporting of all movements.

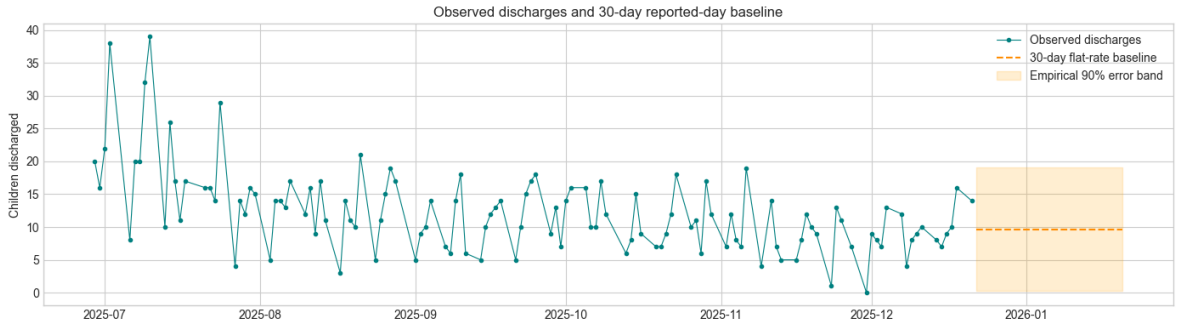


Figure 8: Reported discharge observations and the flat 30-day baseline. The baseline uses recent reported observations; because reporting dates are incomplete, it should not be interpreted as a validated forecast of total calendar-day placements.

9 Streamlit Application and Decision-Support Workflow

The project is accompanied by a deployed Streamlit application, available at the project dashboard. The application is the user-facing companion to the analysis notebook and is intended to make the forecasting workflow and its outputs accessible without requiring the reader to execute the notebook manually. The analytical results described here are grounded in the executed notebook; the web interface should be treated as a presentation and exploration layer rather than as independent validation of the models.

The GitHub repository supplied for this submission is available at the project source repository. Its current contents could not be independently fetched by the review environment, so claims about live code behavior are limited to the supplied notebook and project brief.

9.1 Application functionality and reproducibility

The Unified Mentor brief specifies the desired application modules as a future-care-load chart, a discharge-demand panel, model selection and comparison, and uncertainty visualization. It also specifies a forecast-horizon selector, model toggle, and scenario-comparison view. Because the hosted interface and repository source could not be independently retrieved during this review, this report does not claim that every specified control or panel was verified in the live deployment. The numeric model results reported here are from the executed notebook, not a fresh run of the hosted app. For reproducible use, the application should show the input-data cutoff, selected model, forecast horizon, validation design, metric definitions, and run timestamp. The workflow schematic below is explanatory; the report includes no screenshot of the live dashboard.

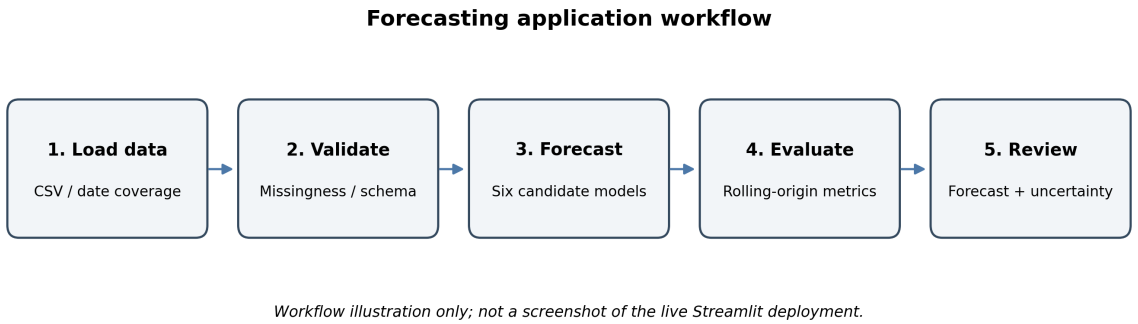


Figure 9: Workflow illustration for the Streamlit forecasting application. This is a generated schematic of the documented analytical workflow, not a screenshot of the live deployment.

The underlying workflow has four principal analytical outputs. First, it summarizes the observed program series and data coverage, distinguishing the HHS care population (a stock) from transfers and discharges (reported flows). Second, it compares six candidate forecasting methods using time-ordered, rolling-origin evaluation. Third, it produces a 30-calendar-day HHS care-load forecast from the final date in the supplied extract and displays empirical error bands. Fourth, it reports a recent observed-record discharge baseline and a net-flow pressure indicator. The results from the notebook run are summarized in tables 4 to 6; they should not be assumed to be the current outputs of a subsequently refreshed application unless the input data, model configuration, and run timestamp are verified.

For reproducible use, the application should display the source-data cutoff, last successful refresh, selected model, forecast horizon, validation metrics, and limitations alongside each forecast. The historical run analyzed in this paper ends on 21 December 2025. Its 22 December 2025–20 January 2026 forecasts are therefore historical experimental outputs, not current estimates. A production-facing deployment should refresh the source data, rerun the time-ordered evaluation, and make these freshness indicators prominent before outputs are used for planning.

The application must also distinguish descriptive indicators from operational alerts. The

recent net-flow pressure baseline of approximately −1.1 children per reported day and the historical mean of −44.7 are summaries of reported flows, not exact changes in the HHS care population. Since the supplied data contain no approved capacity threshold or facility-level staffed-capacity measure, the application cannot substantiate an actual capacity-breach alert from this dataset alone. Any future alert should require a verified capacity input, calibrated uncertainty, and review by authorized program stakeholders.

10 Early-Warning and Capacity-Planning Framework

10.1 Capacity is an external operational input

A forecast becomes a capacity warning only when it is compared with a verified capacity measure. The supplied CSV contains no approved capacity threshold, staffed-bed count, facility-level capacity, medical utilization, or caseworker workload threshold. In the notebook, the capacity variable is deliberately unset. Consequently, no actual capacity-breach date, breach probability, or surge lead time can be estimated from this extract alone.

Let C_t be a verified capacity threshold for the relevant operational unit. A simple rule could flag a potential warning when the upper forecast range crosses capacity:

$$U_{t+h} \geq C_{t+h}. \tag{30}$$

The first forecast crossing is

$$h^* = \min\{h \in \{1, \ldots, 30\} : U_{t+h} \geq C_{t+h}\}. \tag{31}$$

If no such h exists, no crossing is identified within the forecast horizon. This is a rule-based flag, not a probability of overcrowding. It also depends on the capacity threshold being current, approved, and relevant to the population being forecast.

10.2 A probabilistic formulation for future work

If calibrated predictive distributions become available, the probability of exceeding capacity can be defined as

$$p_{t+h} = \Pr(H_{t+h} \geq C_{t+h} \mid \mathcal{I}_t), \tag{32}$$

where $\mathcal{I}_t$ denotes the information available at forecast time t. This probability cannot be derived reliably from the current empirical bands alone. It would require calibrated predictive distributions or a simulation procedure whose coverage and assumptions have been tested.

A practical warning system could use multiple levels, with thresholds agreed by program owners. For example, a preparation warning could be triggered when an upper forecast quantile approaches capacity, while a higher-level alert could require a higher

estimated probability of exceeding capacity. Thresholds should be set through operational consultation, not chosen merely because they make an attractive dashboard.

Table 7: Illustrative alert logic for a future operational implementation. These are design concepts, not active thresholds in the current notebook.

Status	Possible rule	Interpretation
Routine	Forecast range remains comfortably below verified capacity	Continue routine monitoring.
Prepare	Upper forecast bound approaches or crosses capacity	Review staffing, beds, case-management capacity, and placement coordination.
Urgent review	Calibrated breach probability exceeds an approved threshold, or verified current load is already above capacity	Escalate to authorized operational decision-makers.

11 Discussion

11.1 Main findings

Three findings stand out. First, SARIMA is the best-performing method among the six tested approaches under the notebook's validation setup. Second, its MAE rises from 18.47 children in the first week to 103.73 children over days 15–30, demonstrating that a model's short-term performance should not be assumed to persist at longer horizons. Third, discharge activity can only be summarized cautiously because the flow data have substantial reporting gaps.

These results support the use of a simple statistical model as a promising benchmark for this specific historical extract. They do not establish that SARIMA is universally superior, that its error distribution will remain stable, or that its forecasts are suitable for high-stakes operational decisions without additional validation.

11.2 Why the machine-learning models may underperform

The random forest and gradient boosting models do not outperform SARIMA in this experiment. Several factors may contribute:

- The effective sample is modest for flexible models, particularly after lag features and observed-target restrictions are applied.
- Missing dates require interpolation for feature construction, which can smooth or distort local temporal patterns.

- Recursive forecasting feeds predicted values into later feature vectors, potentially accumulating error.
- Calendar variables may encode reporting regularities rather than stable operational drivers.
- Model hyperparameters were specified rather than exhaustively tuned through nested time-series validation.

These are plausible explanations rather than proven causes. Future experiments should use time-ordered tuning, compare against stronger seasonal-naïve baselines where appropriate, and inspect residuals and error stability.

11.3 The importance of the data-generating process

The absence of Saturday observations is not a cosmetic data issue. It creates uncertainty about whether the dataset reflects actual zero activity, a reporting-system convention, a data extraction process, or some combination of these. The reason must be established with the data owner. If reporting is structurally absent on Saturdays, a model designed around a complete daily series may need a different observation model or a carefully justified calendar treatment.

Similarly, linear interpolation is a modeling assumption. It allows regular-grid models to run, but it cannot recover missing peaks and troughs. Sensitivity analysis should compare the current strategy with alternatives such as models designed for irregular observations, explicit missingness indicators, or analyses restricted to periods with adequate coverage.

11.4 Forecast accuracy is not operational readiness

A forecast with a low average error can still be unsafe for capacity decisions if it occasionally misses large surges, if its error distribution changes during policy shifts, or if the capacity threshold is incorrectly specified. Operational readiness therefore requires more than a leaderboard of MAE values. It requires current inputs, calibrated uncertainty, defined escalation rules, and human review.

The dashboard should clearly distinguish observed values, interpolated inputs, point forecasts, empirical error bands, and capacity thresholds. It should display the training-data cutoff and the date the forecast was generated. If the source data are stale, the interface should explicitly mark the output as historical rather than silently implying that it is current.

12 Limitations

The study has several limitations.

1. **Incomplete calendar coverage.** The extract has 355 missing dates over a 1,075-day span, and no Saturday records. The missingness mechanism is unknown.

2. **Interpolation assumptions.** HHS stock values are linearly interpolated to construct a daily model-input series. Interpolation may suppress genuine changes between observations.

3. **Limited validation origins.** Seven origins provide a small basis for assessing stability. Overlapping windows mean forecast errors are not independent.

4. **No external predictors.** The care-load models primarily use lagged HHS history and calendar features; transfer and discharge forecasts are not integrated into the primary HHS prediction.

5. **Uncalibrated empirical bands.** The error-quantile bands are descriptive planning ranges, not formal prediction intervals with demonstrated coverage.

6. **Preliminary discharge baseline.** The recent mean is based on reported observations rather than confirmed complete calendar days. It is not a validated estimate of cumulative placement demand.

7. **No verified capacity data.** The dataset does not contain approved capacity thresholds or staffing and acuity measures. Capacity-breach probabilities and surge lead times are therefore unavailable.

8. **Historical cutoff.** The source ends on 21 December 2025. The generated forecast ends on 20 January 2026 and is historical output, not a current forecast.

9. **No causal interpretation.** The models forecast temporal patterns; they do not estimate the causal effects of policy, border activity, or humanitarian events.

13 Recommendations and Future Work

The following work is recommended before operational deployment.

13.1 Data governance and freshness

1. Refresh the dataset through the latest authorized reporting date and record the source, extraction time, and data version.

2. Confirm with the data owner whether missing dates indicate non-reporting, suppressed values, or genuine zero activity.

3. Investigate the missing Saturdays and long reporting gaps before interpreting weekly seasonality.

4. Establish consistent definitions and reporting windows for transfers, discharges, and the HHS stock.

13.2 Model evaluation

1. Add more rolling origins and preserve a final untouched test period.
2. Compare against persistence, moving-average, and suitable seasonal-naïve baselines.
3. Tune hyperparameters only within time-ordered training/validation windows.
4. Inspect residual autocorrelation, bias, errors by season, and performance during periods of rapid change.
5. Report MAE and RMSE in children alongside the number of scored cases and the forecast horizon. Avoid describing $100(1 - \text{MAPE})$ as a conventional accuracy percentage.
6. Evaluate prediction-interval coverage on held-out forecasts before labeling any band as a 90% prediction interval.

13.3 Flow modeling and capacity integration

1. Once flow reporting is sufficiently complete, develop dedicated transfer and discharge models with explicit calendar-day semantics.
2. Assess whether forecasts of transfers and discharges improve HHS stock predictions without using unavailable future information.
3. Obtain current, approved capacity thresholds, staffed-bed availability, facility-level capacity, and relevant workload measures.
4. Design alert thresholds with operational stakeholders and test false-alarm and missed-warning rates retrospectively.
5. Keep human decision-makers responsible for interpreting alerts and choosing responses.

13.4 Suggested extended modeling framework

A future model could combine stock dynamics and flow forecasts:

$$\widehat{H}_{t+h} = \widehat{H}_{t+h-1} + \widehat{T}_{t+h} - \widehat{D}_{t+h} + \widehat{U}_{t+h}, \tag{33}$$

$$\widehat{T}_{t+h} = f_T(\mathcal{I}_t, h), \tag{34}$$

$$\widehat{D}_{t+h} = f_D(\mathcal{I}_t, h). \tag{35}$$

This structure would connect intake and exit dynamics to the stock forecast. It is not implemented in the current notebook and would require aligned, sufficiently complete flow

observations, a treatment of unknown movements, and careful uncertainty propagation. If forecasts are generated separately, the uncertainty of both flow forecasts and the stock starting value must be carried into the resulting care-load distribution.

14 Ethical and Responsible Use

The analysis concerns aggregate care-system planning. Forecasts should not be used to make automated decisions about individual children or to substitute for professional judgment. Errors can have asymmetric consequences: underprediction may leave resources inadequate, while overprediction can cause inefficient allocation. Decision thresholds should therefore reflect the actual costs and risks of false negatives and false positives, as determined by responsible program authorities.

The dashboard should communicate uncertainty and data freshness prominently. It should not present empirical error bands as guarantees, nor should it label a system as overcrowded solely because a point forecast crosses an arbitrary threshold. Any operational use should undergo data governance, privacy, security, and program-owner review.

15 Conclusion

This project develops an exploratory forecasting pipeline for the HHS care population and a cautious baseline for observed discharge activity. It compares six approaches under a rolling-origin design and finds that SARIMA produces the lowest MAE among the tested methods for the short, medium, and longer forecast bands. Its MAE increases with lead time, from 18.47 children for days 1–7 to 103.73 for days 15–30. The discharge analysis yields a recent average of 9.6 children per reported day, but incomplete reporting prevents this value from being treated as a validated daily or cumulative placement forecast.

The most important conclusion is methodological: forecast quality depends on the reliability and interpretation of the data, not just the choice of algorithm. The missing dates, absent Saturday observations, small number of rolling origins, uncalibrated error bands, stale forecast cutoff, and lack of approved capacity thresholds all limit operational claims. With refreshed data, stronger validation, explicit flow modeling, calibrated uncertainty, and verified capacity information, the prototype could be developed into a more useful decision-support tool. Until then, it should be treated as an exploratory model comparison rather than an operational warning system.

Data, Code, and Application Links

The analysis was performed in a Jupyter notebook titled *Predictive Forecasting of Care Load & Placement Demand*. The source data file is `HHS_Unaccompanied_Alien_Children_Program.csv`.

The notebook and data extract should be distributed only in accordance with the data owner's access and sharing requirements. Results in this report correspond to the supplied extract ending 21 December 2025 and are not current forecasts. The official HHS program dataset is published on HealthData.gov and was listed as updated in October 2026 (U.S. Department of Health and Human Services, 2026); the report therefore uses a historical project snapshot rather than the latest available data. The data should be refreshed and the analysis rerun if the mentor expects current-date forecasts.

Acknowledgement

The project scope and acceptance criteria were supplied in the Unified Mentor project brief. The dataset is attributed to the U.S. Department of Health and Human Services / Office of Refugee Resettlement as the public data provider.

Project Attribution and Scope

This document is a project report prepared in response to the project brief supplied by Unified Mentor. It uses aggregate public program data attributed to HHS / ORR; it is not an official HHS analysis, an HHS-developed tool, or an endorsement by HHS.

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A Reproducibility Notes

The recorded experiment used Python with pandas, NumPy, Matplotlib, scikit-learn, and statsmodels. The notebook specifies the following principal model settings:

- Exponential smoothing: additive trend, damped trend, additive weekly seasonality, seasonal period 7, optimized fit.
- SARIMA: order $(1, 1, 1)$, seasonal order $(1, 1, 1, 7)$, relaxed stationarity and invertibility constraints, maximum 150 iterations.
- Random forest: 250 trees, minimum leaf size 3, maximum feature fraction 0.9, random seed 42.
- Gradient boosting: 150 estimators, learning rate 0.03, maximum depth 2, Huber loss, random seed 42.
- Validation: final 20% of observed HHS records, origins spaced 28 days apart, 30-day forecast at each origin, scoring only observed future HHS targets.

B Figure provenance and interpretation

Five empirical time-series plots supplied with the project materials are incorporated in the report: observed program measures, the observed/interpolated HHS series, additive decomposition, the historical SARIMA forecast, and the discharge baseline. Additional

diagrams were generated for this report to explain the stock–flow framing, rolling-origin evaluation, and application workflow. The model-comparison chart was generated from the MAE values in the executed notebook's result tables. Generated diagrams are explanatory; they are not independent empirical observations. The live application was not independently inspected, and no screenshot of its interface was available in the supplied materials.